Kunal Roy Choudhury

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Experience

Lead Software Engineer March 2026 – Present

FinStocks AI (first employee, $3M seed) *Bengaluru, India*

- Led a 3-engineer team and shipped a **LangGraph** agent for live Indian-market chat, charting, research and orders; collapsed a 3-stage router into one tool-call node, removing a serial model call and its latency from every turn.
- Architected the natural-language-to-strategy **backtesting engine**: LLM output is a typed Pydantic IR (185 indicators, 148 patterns), statically validated and compiled to Numba-JIT kernels that backtest the full universe over float32 Parquet; the same IR and compiler drive live execution, with tick coalescing and Redis Lua for atomic capital allocation.
- Kept the model away from the money: a 1.4K-line validator enforces provenance for every level on an LLM chart terminal (85 typed operations), and `place_order` takes no arguments and rejects mismatched confirmations.
- Built the eval platform: 73 prompts, 18 model configs, cost and TTFT lineage, and a CI gate fed by production failures.
- Cut market-data queries from **13.6 s to 48–135 ms** (and 5.9 s to 782 ms) with a Postgres job queue (`SKIP LOCKED`, visibility leases, idempotent partial indexes); one Redis-fed tick stream serves every WebSocket on GCP Cloud Run.

Software Developer April 2025 – February 2026

Indian Kanoon *Bengaluru, India*

- Sole engineer on Prism: designed, built and launched Indian Kanoon's AI legal-research suite (Django, PostgreSQL, Redis, Celery, Gemini) into a platform with **1.5M+ users** and 10+ LLM tools, then scaled it to 5,000+ concurrent users.
- Cut median RAG latency by **40%** and reached **92%** grounded query accuracy over millions of court judgments, using Postgres full-text search, query expansion, multi-level caching, index tuning and citation grounding.
- Built LLM workflows for legal Q&A, drafting, case prediction and counter-arguments, with citation grounding and evals.

Software Engineer Intern September 2024 – April 2025

Chargebee *Chennai, India*

- Migrated **500K+ subscriptions** of enterprise billing data with zero customer-facing downtime in a large-scale database optimization effort, using a Java framework built on JOOQ schedulers (PostgreSQL) with Vue.js validation tooling.

Software Engineer Intern January 2024 – September 2024

AiDash *Bengaluru, India*

- Helped cut the release cycle by **60%** by extracting services from a 2M+ LOC monolith (Spring Boot, hexagonal architecture); optimized SQL pagination over 100K+ record tables across Django, MongoDB and S3 services.

Open Source

- **23 merged PRs across 11 projects** (all PRs): Turso (25K stars), OpenObserve (23K, 2), GitButler (22K), Scalar (16K, 7), Trail (11K, 4), Bifrost (9K), pgstream, Respan Gateway (2), xalen-ephemeris (2), pkg.pr.new, Needle. Also **93 issues filed across 37 projects** (all issues), including hashicorp/raft, SigNoz, NATS Server and KeePassXC.

Projects

Gostman | *Go, React, Wails, REST, GraphQL, gRPC, WebSockets* *live* | *code*

- Open-source API client (desktop, web, CLI; **81 GitHub stars**): **<1s cold launch**, **90% lower memory** than Electron. `gostman run` for CI; `gostman mcp` lets coding agents call saved requests without seeing secrets.

AeroSieve | *Rust, WebAssembly, WebSockets, Python bindings* *live* | *code*

- Speech-data curation pipeline: curates 43 min of Hindi audio in **2.75 s** on a laptop CPU (**∼940× real time**); caught 24,385 of 24,409 injected defects across six languages, and holds 1,000 live streams at 14.8 ms p95.

Seedfall | *Go, wazero, deterministic simulation* *live* | *code*

- Replays any failure of an unmodified Go program from one seed; found hashicorp/raft never fsyncs snapshots (raft#744).

Technical Skills

Languages & Frontend: Python, TypeScript, Go, Rust, Java, SQL, React, Next.js, React Native, Vue.js
Backend & Data: Django, FastAPI, Spring Boot, gRPC, GraphQL, Celery, PostgreSQL, Redis, MongoDB, Elasticsearch
Cloud & AI: AWS, GCP, Docker, Kubernetes, CI/CD, LangGraph, RAG, evals, Anthropic, OpenAI, Gemini, Claude Code

Education

SRM Institute of Science and Technology Kattankulathur, India

B.Tech, Computer Science and Engineering, ***CGPA: 9.24 / 10*** *August 2020 – September 2024*